

Launch Vehicle Ascent Phase Loading: Comparing Traditional and CFD-Based Approaches

Alexis Castel, Ph.D.
Keysight Technologies

Motivation & Context

- Following Artemis I, NASA/Boeing conducted an extensive study of the SLS ascent environment
- Critical finding: vortex shedding between the SLS core stage and solid rocket boosters produced low-frequency excitation that traditional Corcos TBL models could not capture
- Corcos assumes homogeneous spatial correlations → fails for vortex-dominated, low-frequency, spatially non-uniform phenomena
- Both random and deterministic FSP approaches were used to characterize these environments

The study was a success: comprehensive environment envelopes were derived from multiple trajectory cases

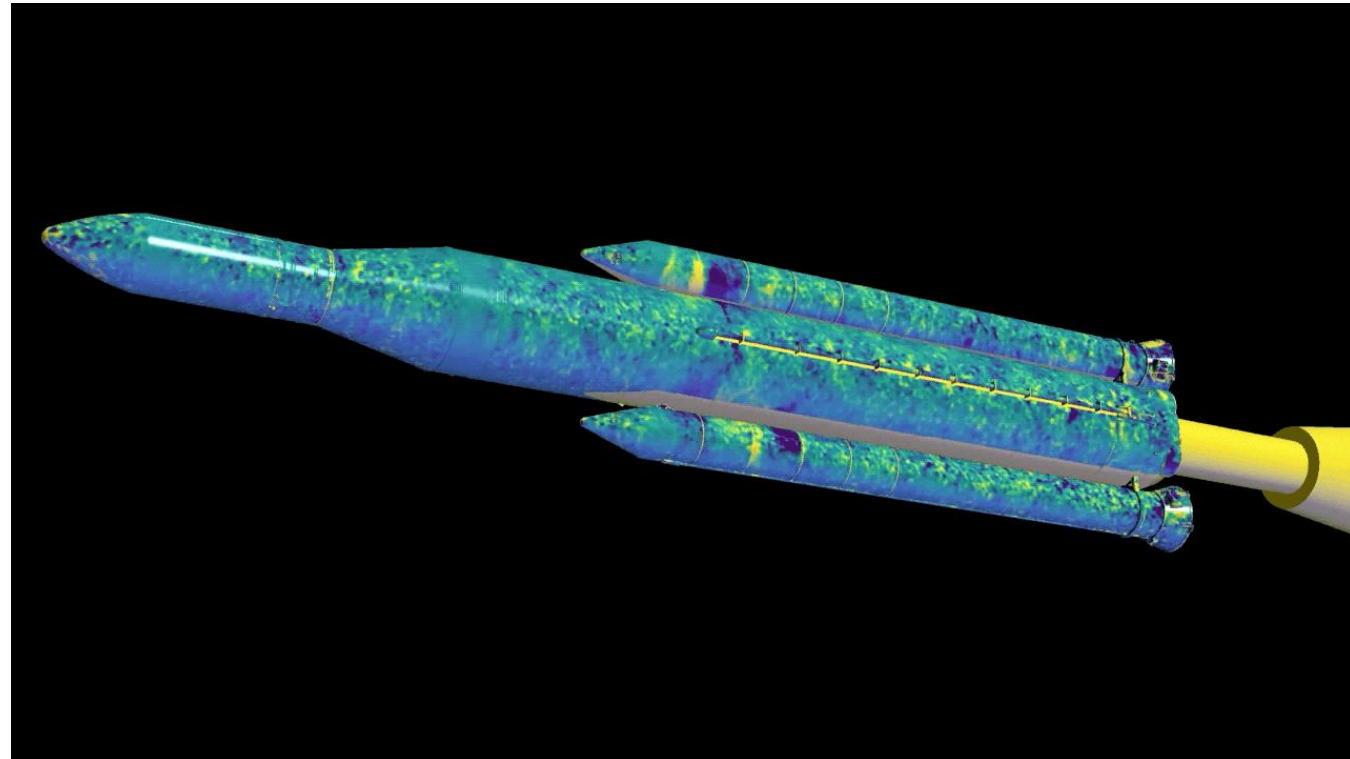


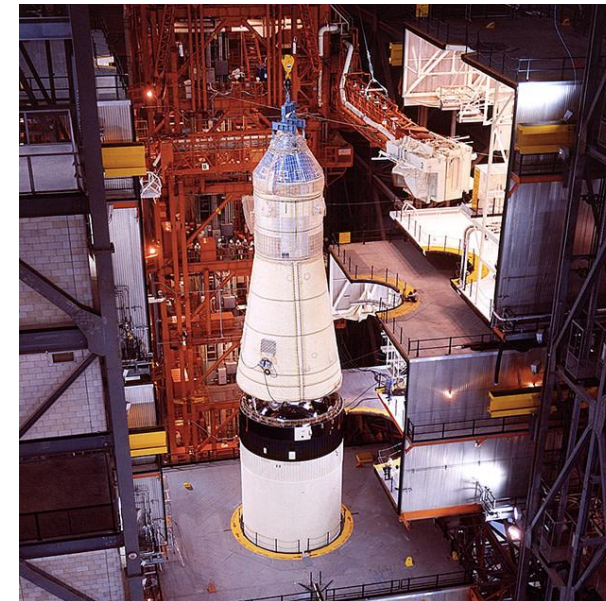
Image credit: nasa.gov

The Problem: Where TBL Assumptions Break Down

- Corcos TBL assumes homogeneous spatial correlations — works well for smooth flat/cylindrical sections
- Breaks down at:
 - Flow separation regions
 - Geometric discontinuities (SLA taper, inter-stage flanges, protuberances)
 - Non-equilibrium boundary layers
- Need to capture local flow variations that empirical fitting cannot

Study Overview & Vehicle Description

- Generic launch vehicle: Saturn V SLA (extensively documented, publicly available geometry)
- Focus on the SLA section: houses the Lunar Module, good candidate for internal acoustics prediction
- CFD simulation of ascent phase: pressure vs. time imported on the SLA surface
- Goal: compare methodologies for predicting internal acoustics within VA One



Methodologies

Using CFD data to load a structure

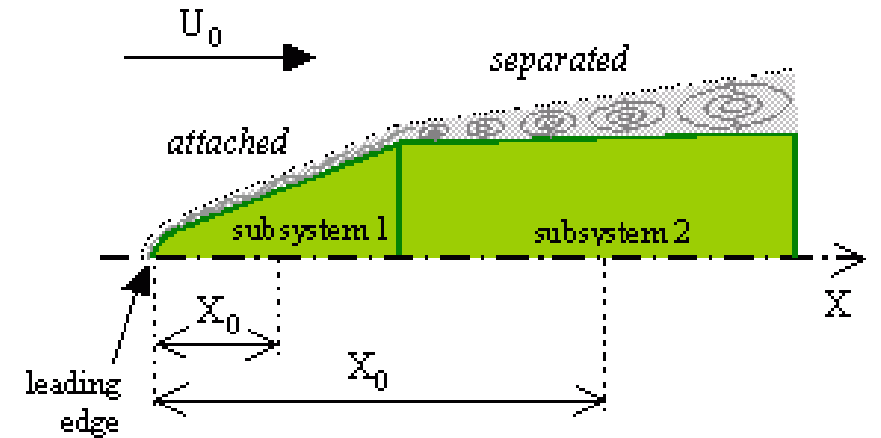
- Three options:

Starting Point	Path	Load Type	Applicable Subsystems	Frequency Range
Wind tunnel data CFD $p(x,t)$	Fit TBL parameters	Corcos TBL (fitted c_x , c_y , U_c)	FE Structure, SEA Structure	Full frequency (analytical)
CFD $p(x,t)$	Direct mapping	FSP (deterministic or random)	FE Structure, BEM Fluid	Low frequency (modal regime)
CFD $p(x,t)$	Wavenumber transform	GSP (WFS + pressure spectrum)	SEA Structure and FE Structure (new!)	Low through high frequency

TBL model only extracts the energy fitting into the Corcos model. If there is an acoustic component or another component, it is not accounted for.

Approach 1: Traditional Corcos TBL

- Analytical model: exponentially decaying spatial correlation in streamwise & cross-flow directions
- Parameters:
 - Convection velocity U_c , decay coefficients c_x, c_y
 - Typically derived from wind tunnel measurements
 - fitted from CFD via the FSP → Fit TBL workflow in VA One
 - Pressure can be analytically calculated or measured (most common)
- Applied as a **random load** on FE or SEA subsystems
- **Limitation**: assumes the pressure field is spatially homogeneous



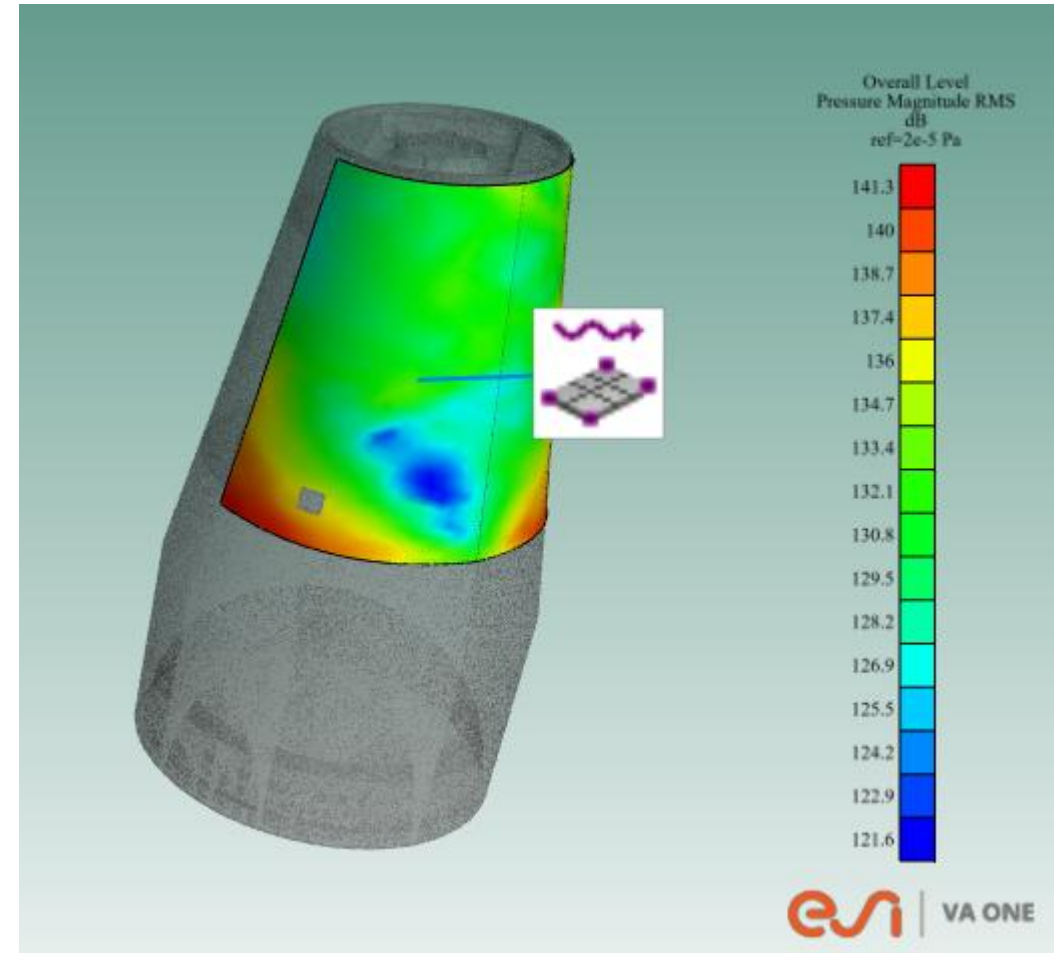
$$R(x, x', \omega) = \exp\left[-\frac{c_x |\Delta x|}{d} - \frac{c_y |\Delta y|}{d}\right] \times \exp[-ik_c \Delta x]$$

$$\frac{1}{d} = k_c \sqrt{1 + \left(\frac{8}{3k_c \delta}\right)^2}$$

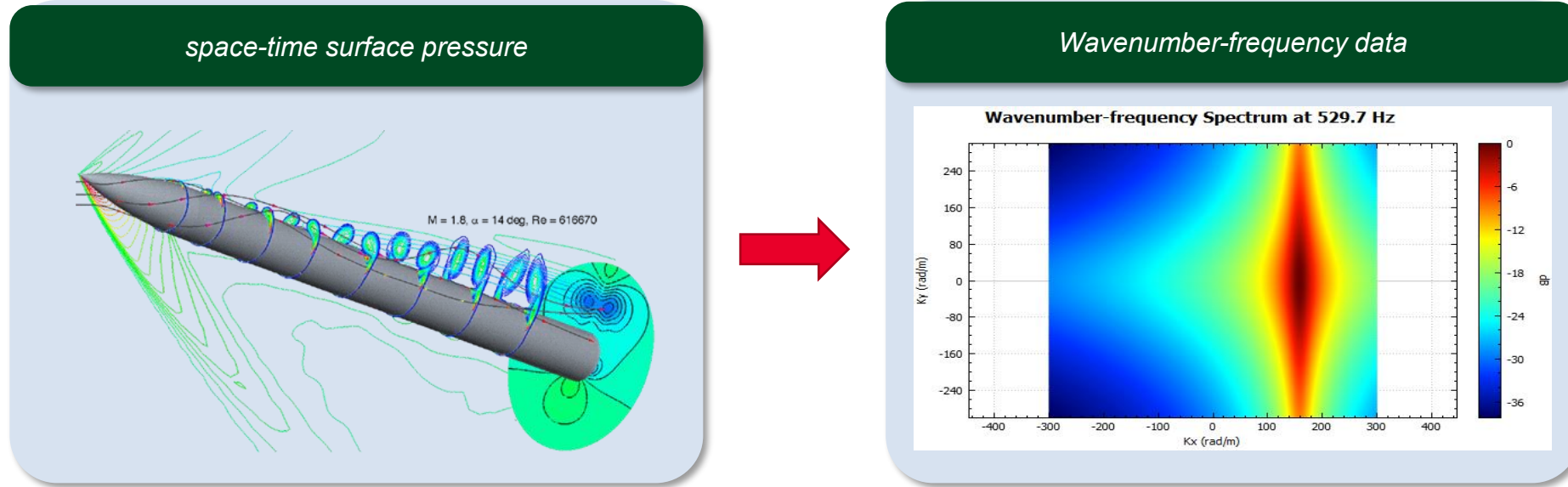
If flow is parallel to x axis, typical decay coefficients could be: $c_x=0.10$, $c_y=0.72$

Approach 2: Fluctuating Surface Pressure (FSP)

- Direct mapping of CFD pressure time histories onto FE structural mesh
- Two modes of application:
 - **Deterministic** (preserves phase → transient peak response)
 - **Random** (statistical envelope)
- Captures spatially-varying pressure fields including local flow effects
- Best suited for **low-frequency modal regime** (below ~200–500 Hz)
- Mesh mapping: consistent vs. conservative — works even with incompatible CFD/FE meshes

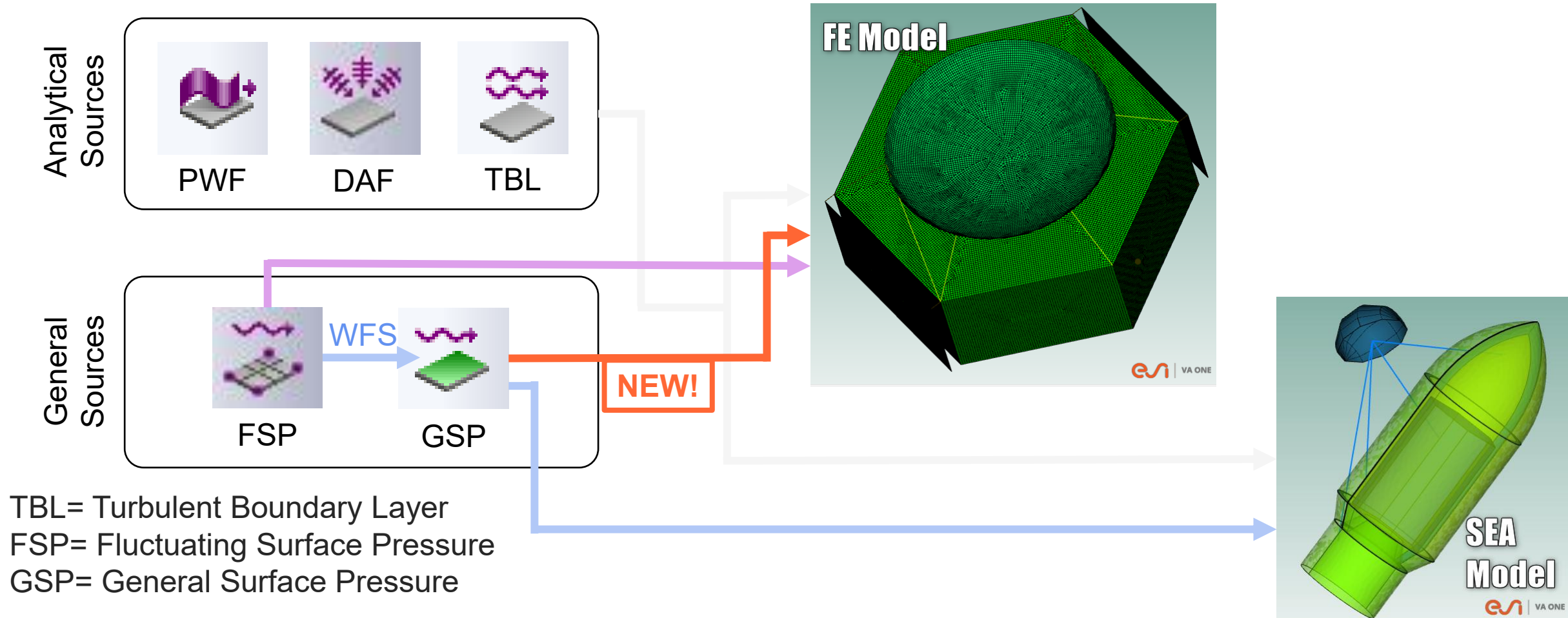


Approach 3: Generic Surface Pressure (GSP)



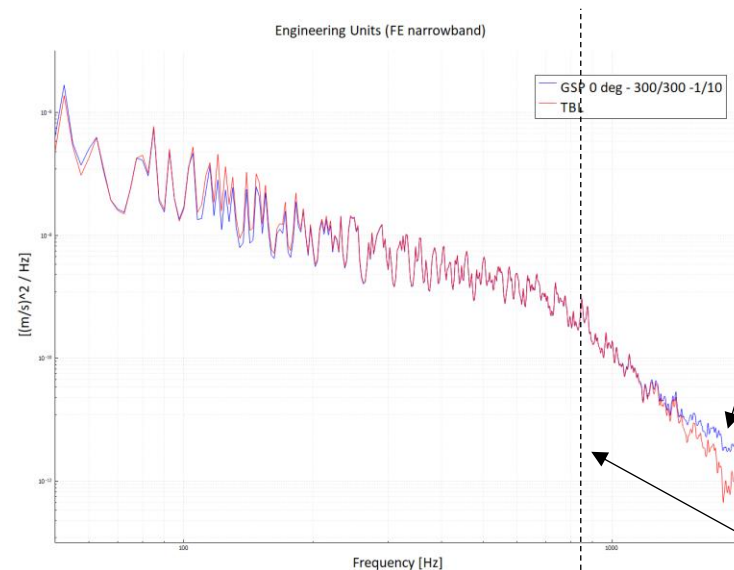
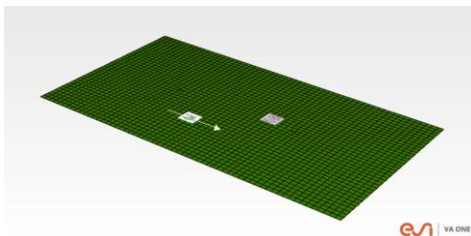
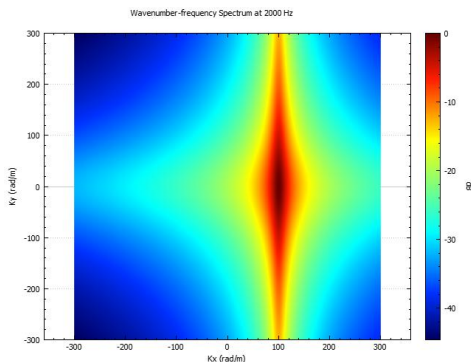
- Wavenumber-frequency transformation of the CFD $p(x,t)$ signal
- Separates energy into:
 - Convective wavenumber components → dominate structural excitation
 - Acoustic wavenumber components → dominate panel transmission (mass law)
- No analytical assumptions — physics-based characterization
- Originally for SEA subsystems only
- **New in VA One 2024: GSP can now also be applied to FE structures → seamless low-through-high-frequency in one framework**

GSP on FE



GSP on FE

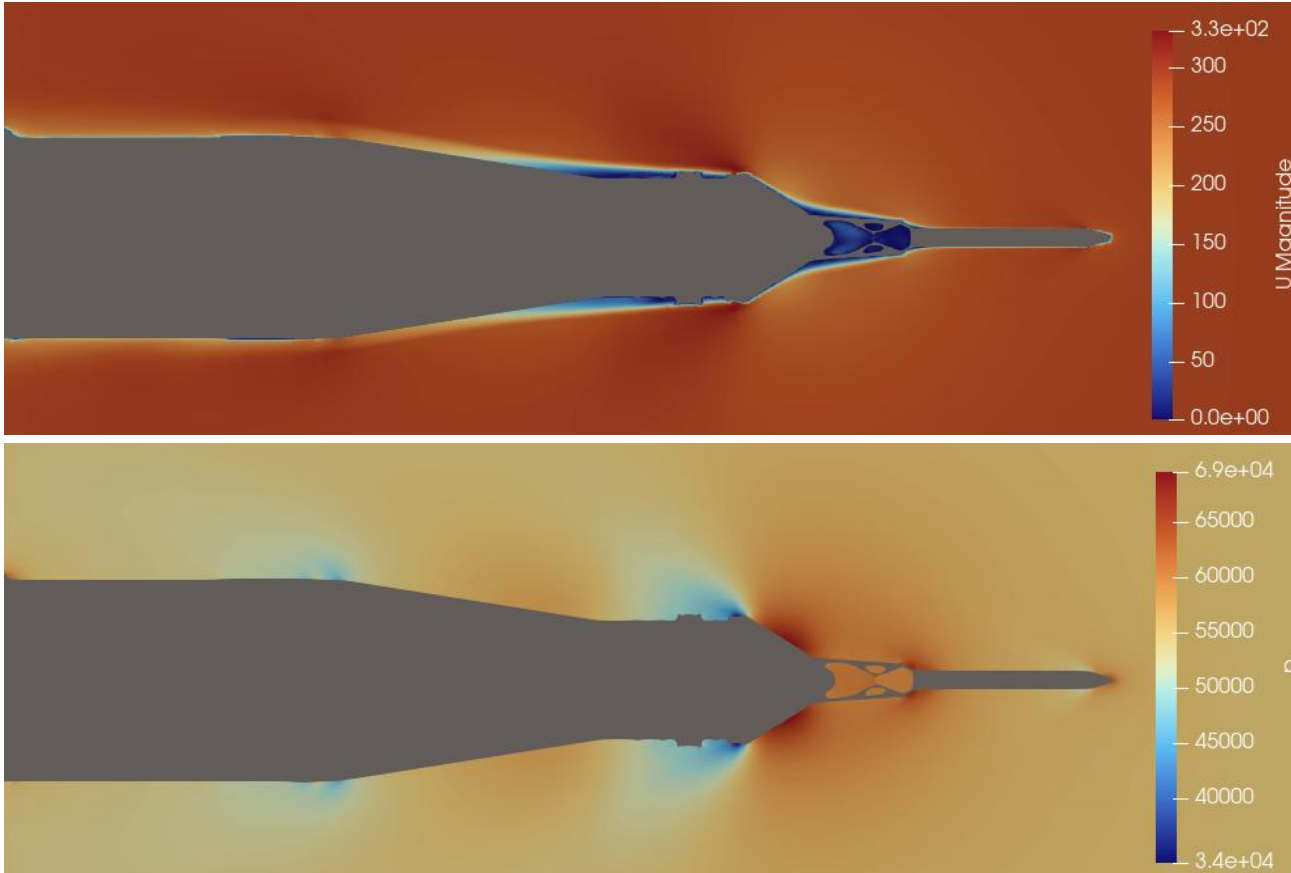
- Guidelines for an FE Application
 - If the **Pressure Distribution** over the excited surface of the structure is **known** (BEM, CFD, test) → **use FSP** for best accuracy
 - If the **Pressure Distribution** over the excited surface of the structure is not known and an approximation is sought (re-use pressure load from another project/model) → use GSP to generate a universal source
- No mesh-too-coarse issues like with FE TBL (wavenumber content is filtered)



No effect on GSP
because wavenumber
content is filtered
according to FE mesh

FE TBL "mesh too
coarse" warning
message → results are
inaccurate > 1 kHz

Saturn V SLA: CFD Model & Data Import

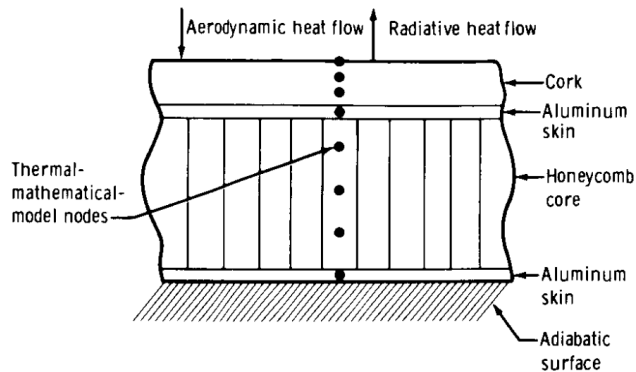
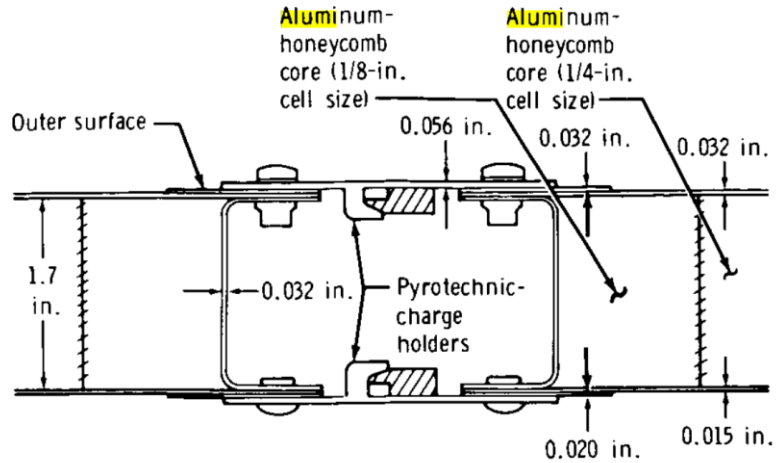


Parameter	Value
Solver	OpenFOAM v2112 — rhoPimpleFoam
Flow regime	Mach 0.9 (transonic, compressible)
Air properties	Atmosphere at 16,000 ft
Turbulence model	k- ω SST (RANS)
Mesh size	~20 M cells, 97.7K wall faces on vehicle
Time stepping	Adaptive, $CFL \leq 0.8$, $\Delta t_{\max} = 10^{-4}$ s
Pressure sampling	$\Delta t = 0.2$ ms on rocket patch
Simulation window	0.4 s of usable signal (after initial transient of 0.1 s)
Output	Wall pressure time series → UNV format
HPC	128 MPI ranks (scotch decomposition)

- Saturn V geometry with the SLA pressure band highlighted (x = 46–58 m)
- Mach contour plot or instantaneous pressure snapshot on the SLA surface

Saturn V SLA: VA One Model Setup

- FE structural model of the SLA (generic geometry with sandwich panels, cork TPS, LM mass)
- FE acoustic cavity (shrinkwrap interior)
- SEA version of the model for high-frequency comparison
- Boundary conditions representative of the vehicle



Apollo flight vehicles SC-009, SC-011, and SC-101 did not include an LM but used a stiffening member in the SLA in place of the LM. This stiffener, shown in figure 14, weighed approximately 70 pounds and provided the stiffness necessary to stabilize the SLA shell structurally.

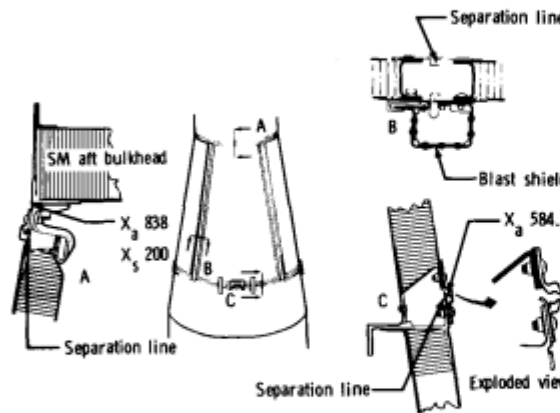


Figure 13. - Adapter panel separation lines.

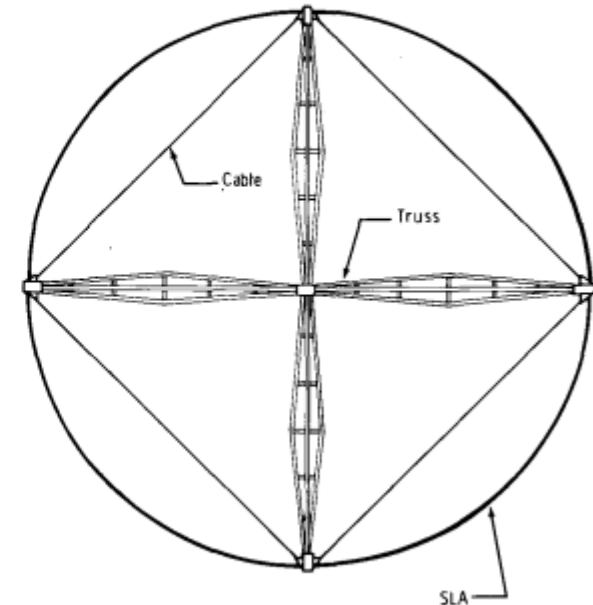
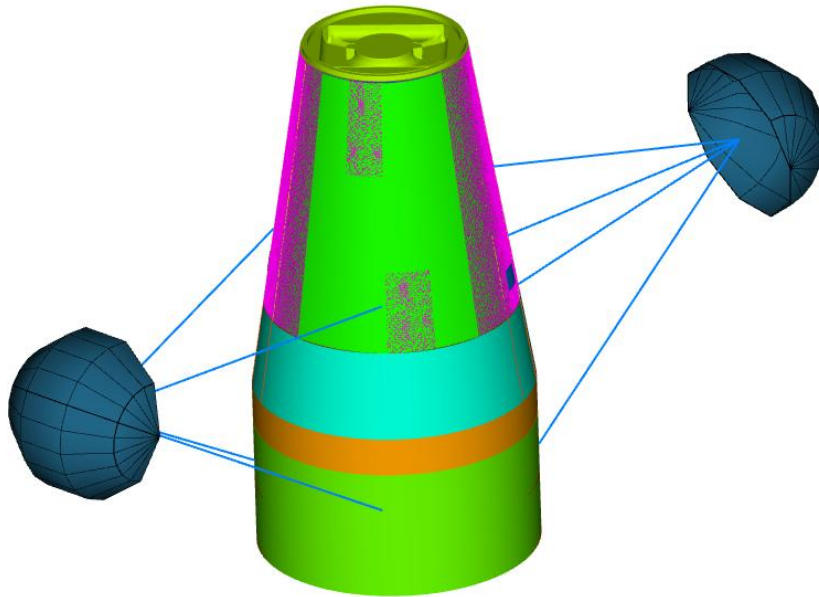


Figure 14. - Structural stiffener.

Saturn V SLA: VA One Model Setup

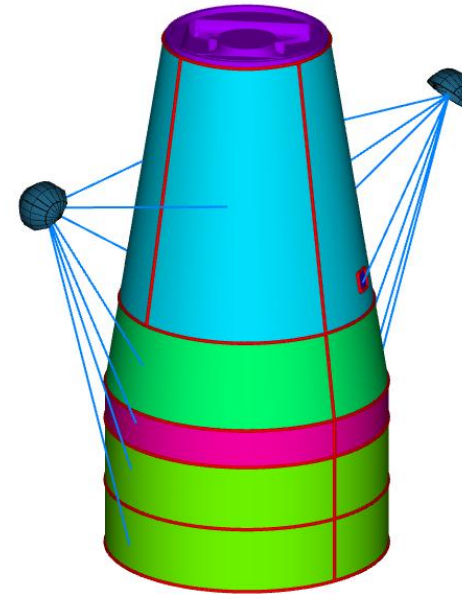
- Equivalent generic material properties are chosen based on documented description.
- Studied response:
 - Cavity average Sound Pressure Level
 - SLA Panel responses



FE Structural subsystems
3434 Modes up to 300 Hz



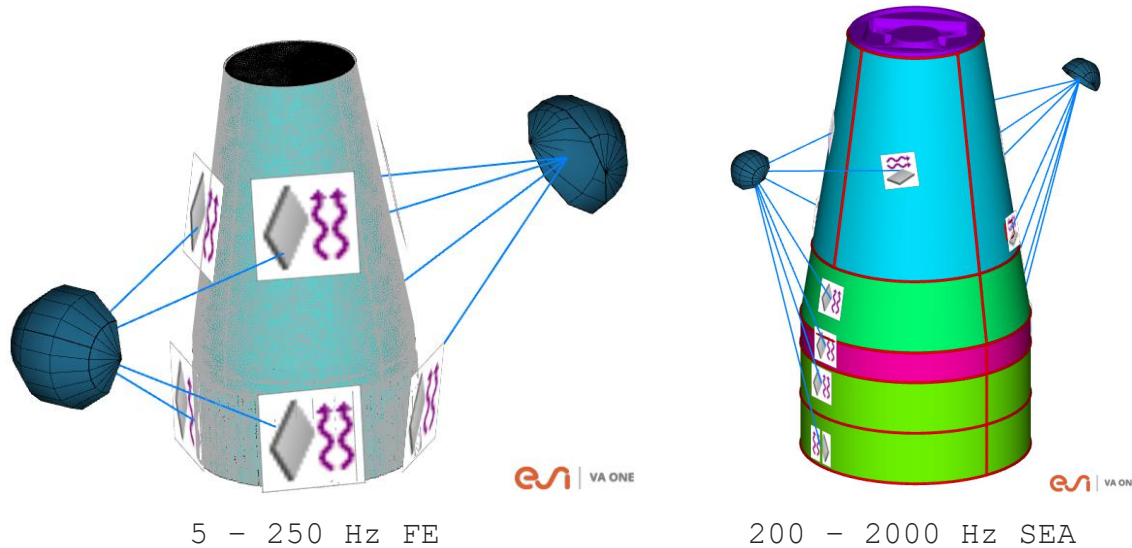
FE Acoustic Cavity
1588 Modes up to 300 Hz



SEA Model
200 Hz+

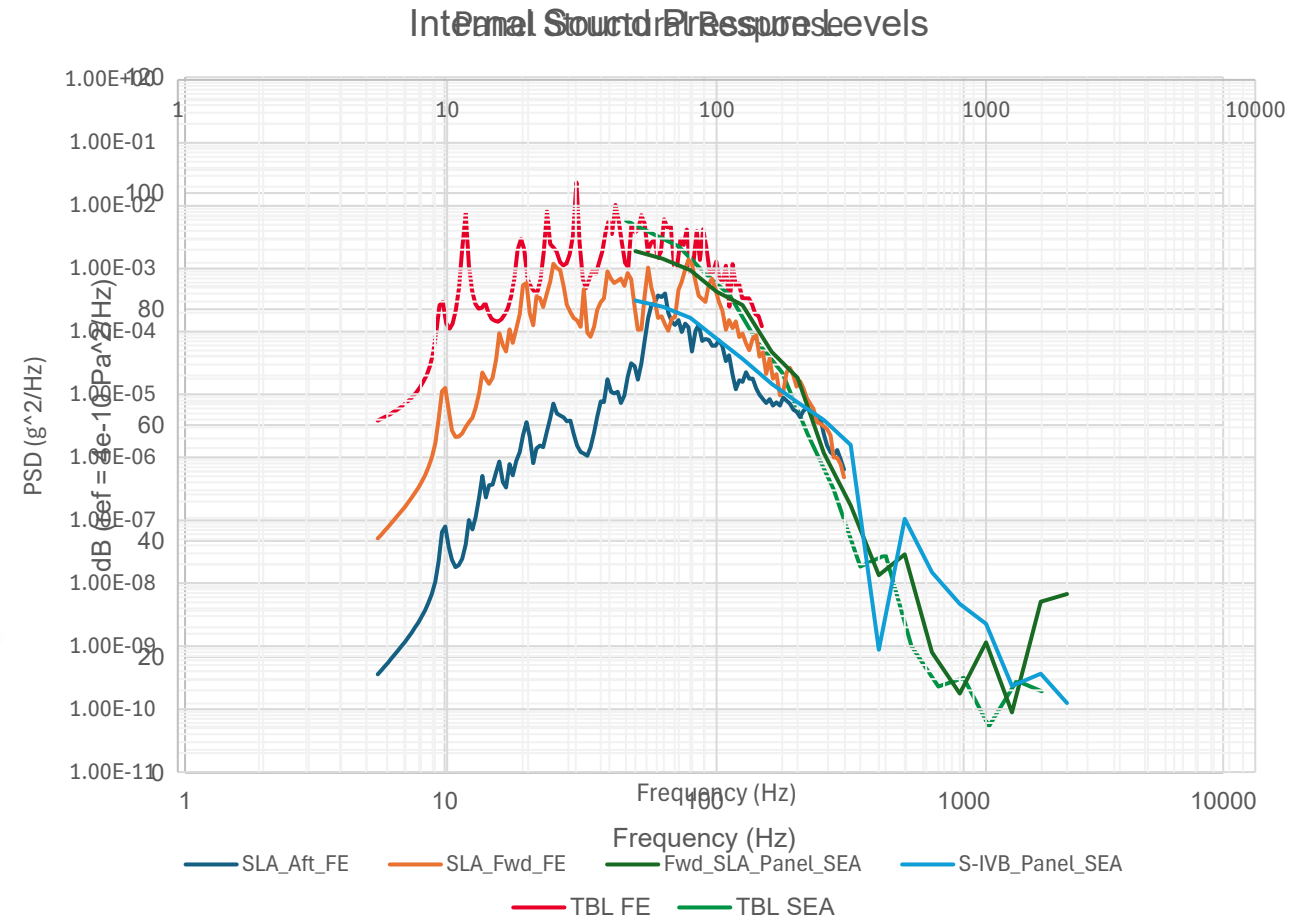
Case 1: TBL Loading on the SLA

Applying the analytical model



- Using FIT TBL parameters

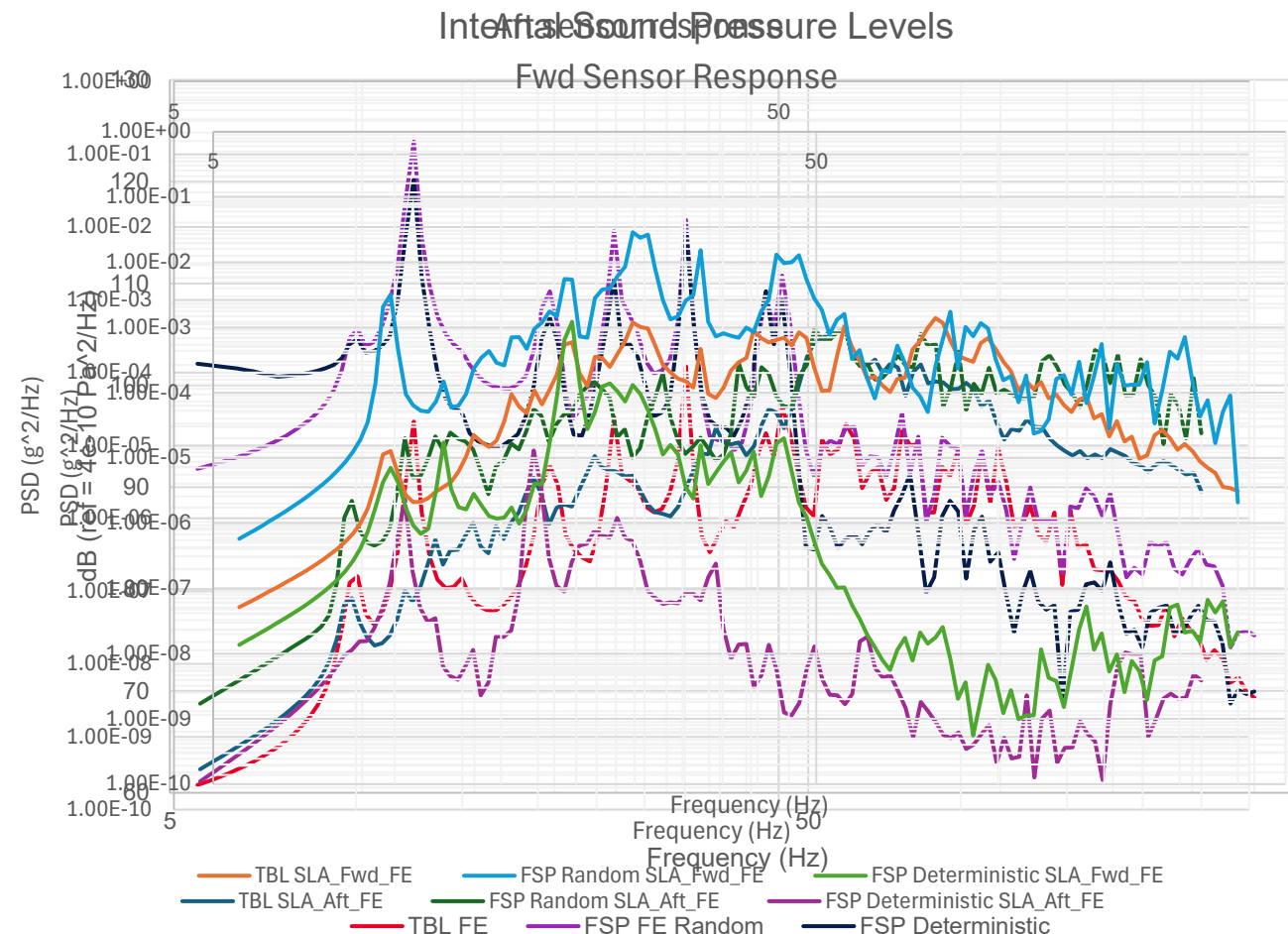
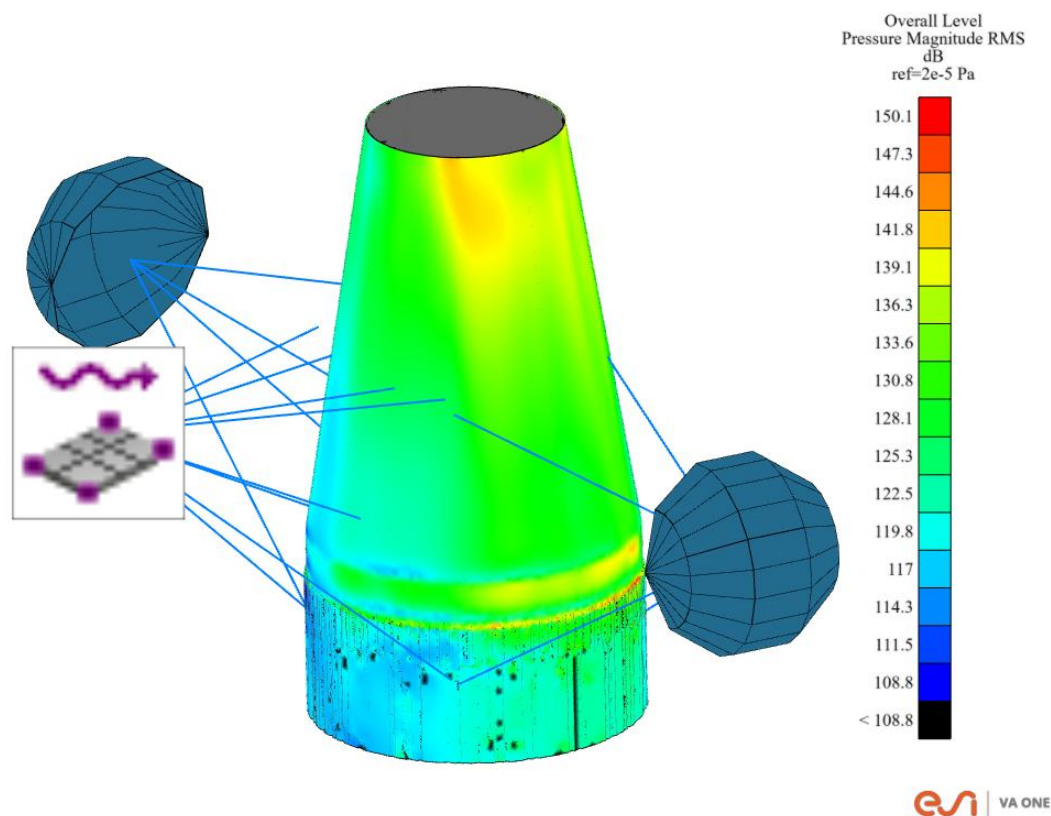
- ➔ VA One reads CFP pressure vs time data to obtain TBL Decay coefficients.
 - ➔ Pressure Spectrum
 - ➔ Convective wavenumber
 - ➔ Decay coefficients



➔ Results show continuity

Case 2: FSP Loading on the SLA

Low frequency only option

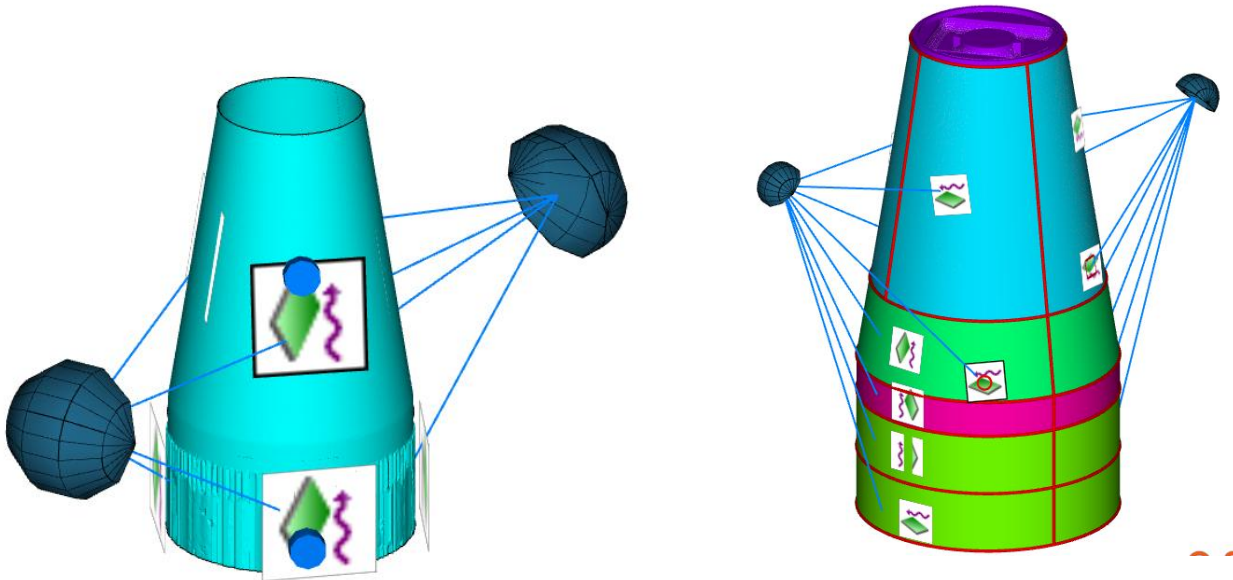


- CFD Pressure is mapped directly on the structure
 - Contains both the TBL content and other information
- Deterministic or random (Welch's process)

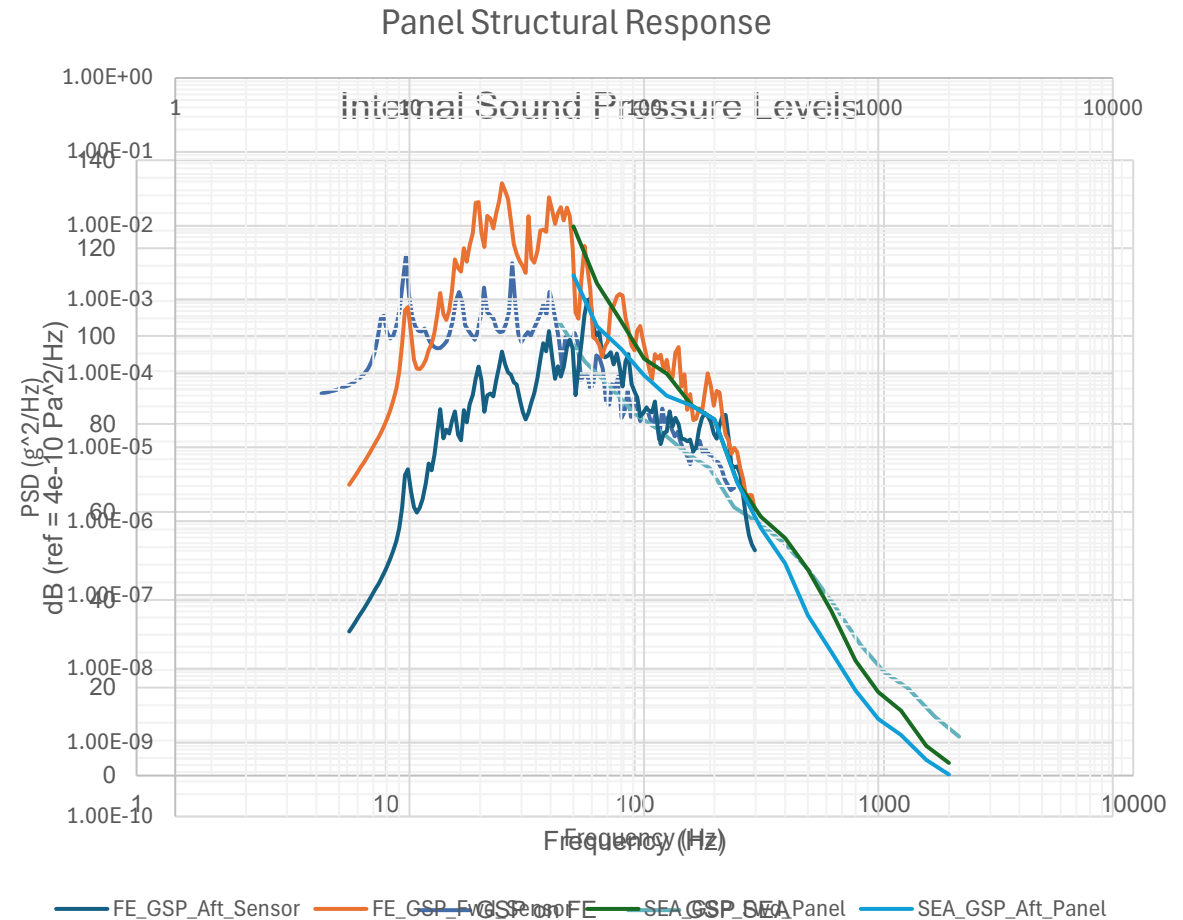
- ➔ Random vs deterministic has an influence on results.
- ➔ The deterministic vs random effect is expected to be more important if multiple vortices are shedding at low frequencies.

Case 3: GSP Loading on the SLA

SEA Results

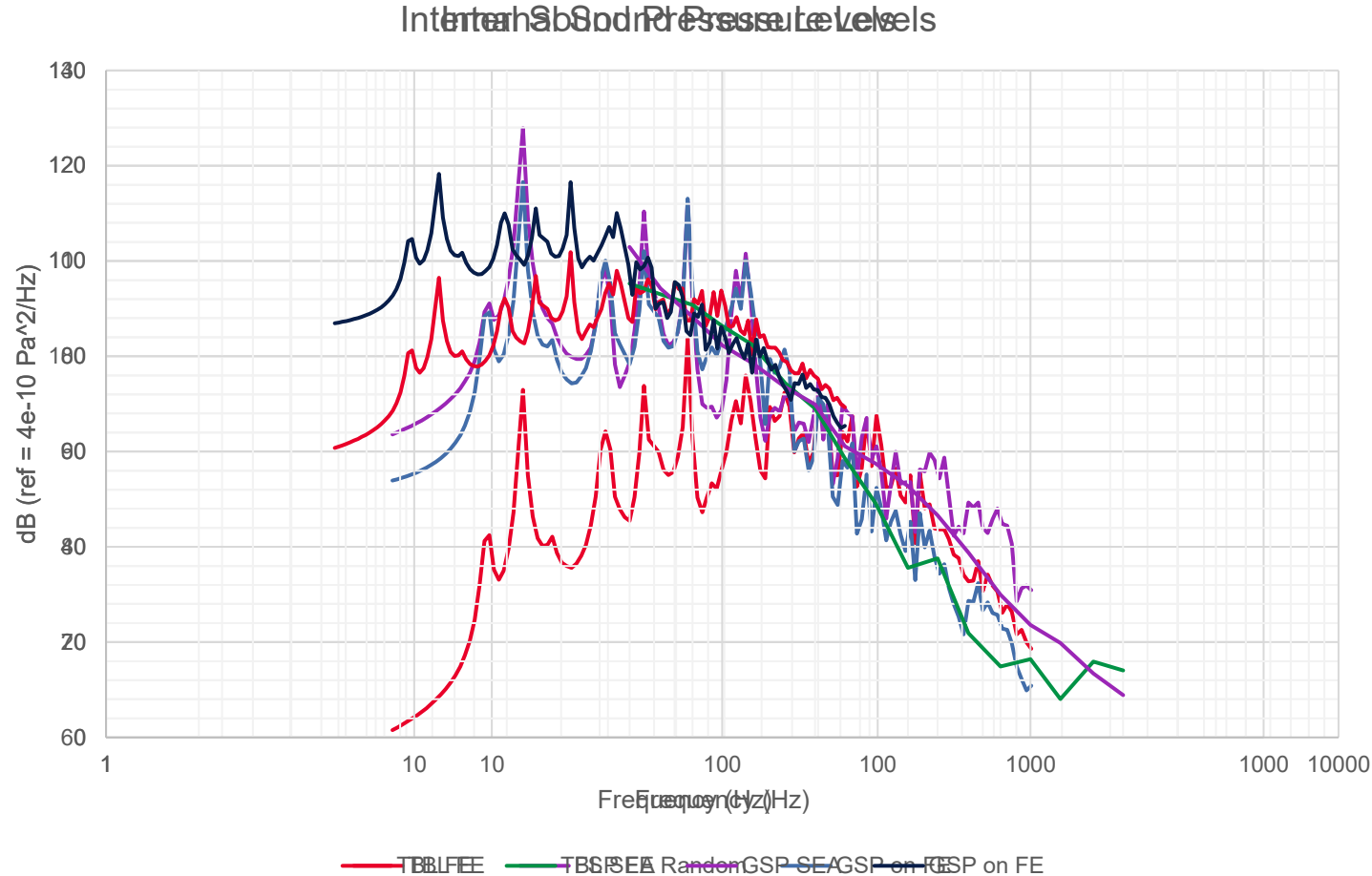


- Wavenumber transform is performed from random sound field (defined by cross-spectral matrix).
- General Surface Pressure (GSP) is applied on box FE and SEA models.



➔ Results show continuity

TBL vs FSP vs GSP



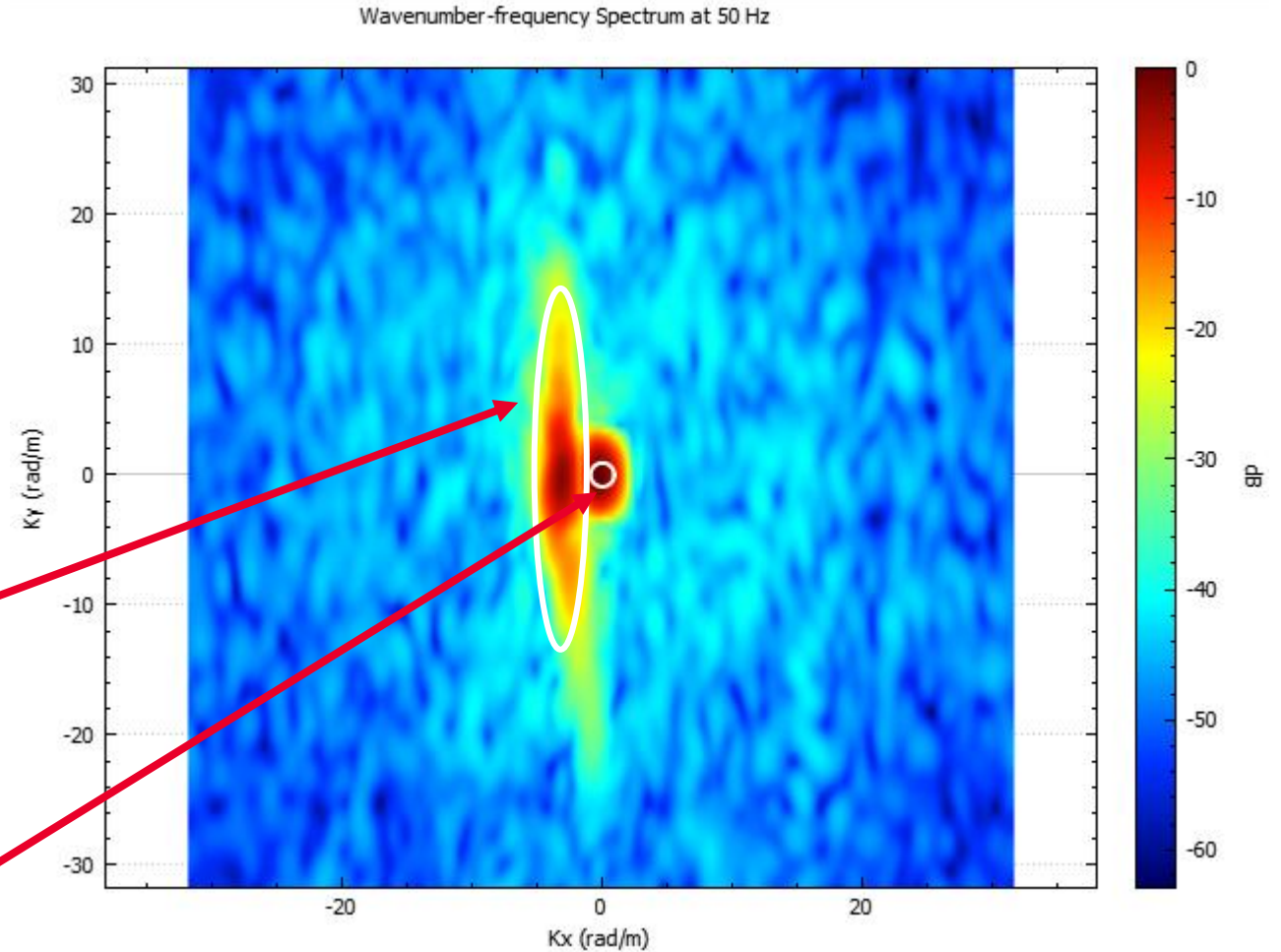
- GSP → **loses local spatial details**
- FSP → **retains local spatial details**
- ➔ **GSP is good approximation for FSP**
- TBL results are quite lower
 - CFD results are also filtered/fitted in to match a TBL
 - Some energy is filtered
- Similar conclusions follow for structural responses

Wavenumber analysis

- Show both a acoustic and hydrodynamic content

Hydrodynamic content

Acoustic energy

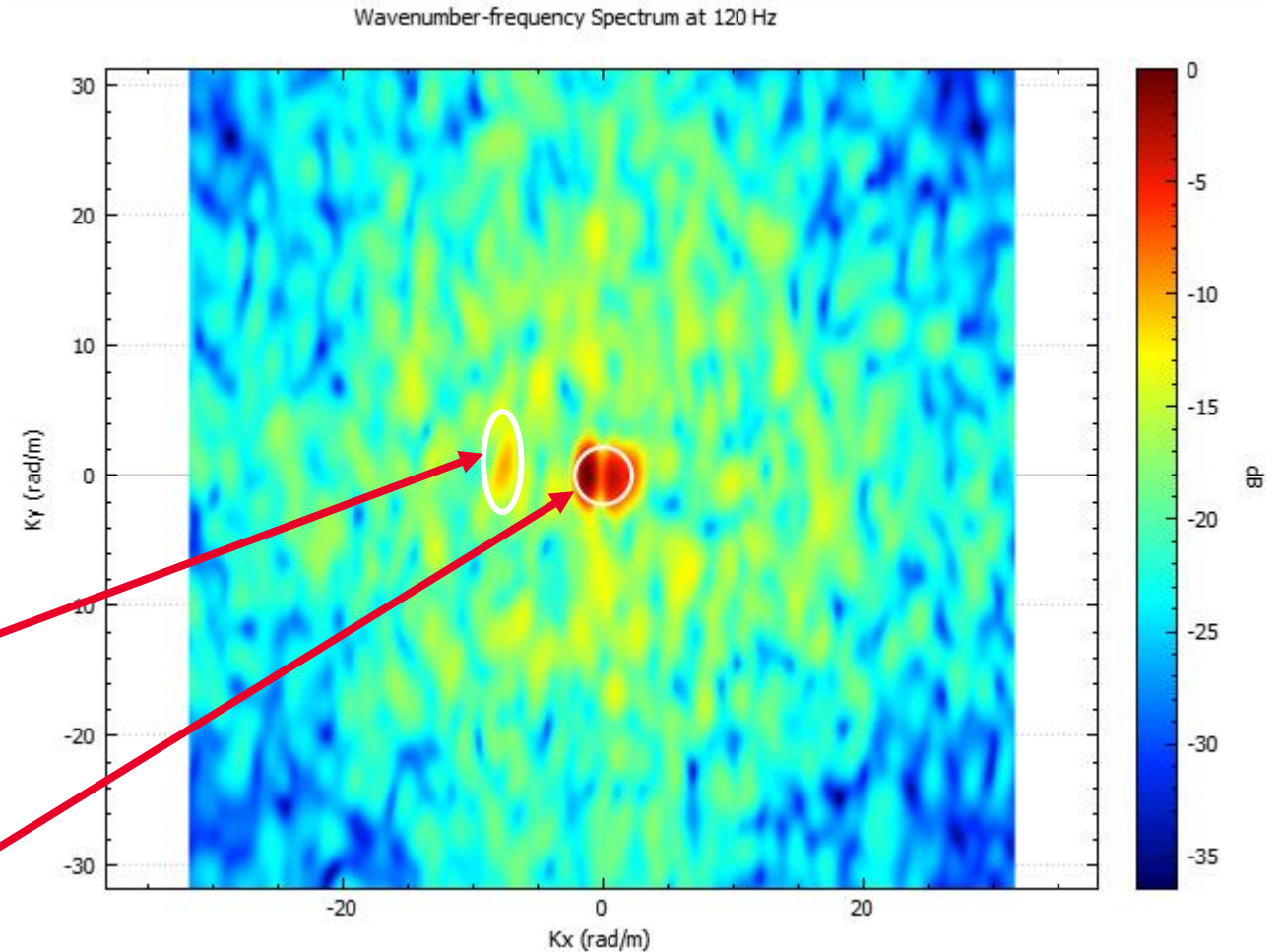


Wavenumber analysis

- At some frequencies the acoustic energy is dominant
→ Here TBL could be a bad fit.

Hydrodynamic content

Acoustic energy

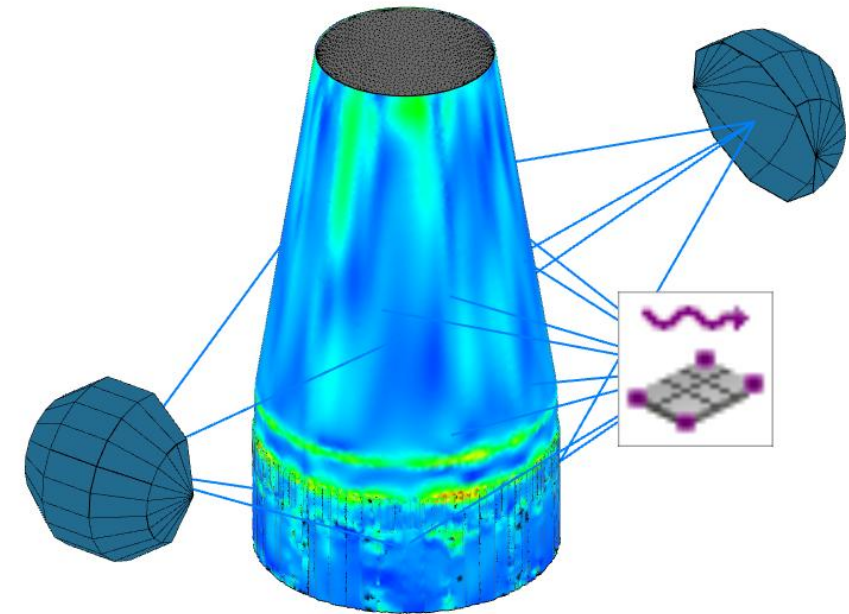


Discussion: When to Use What

Method	Best For	Frequency Range	Key Limitation
TBL 	Smooth surfaces, quick assessment	Full (analytical)	Homogeneous assumption
FSP 	Local effects, phase, deterministic peaks	Low (modal)	Mesh + cost
GSP 	Universal random load, DFAT, mixed acoustic+TBL	Full	Loses phase Data aliasing

Conclusions

- CFD-based loading **changes the predicted physics**, not just accuracy
 - Could be used to apply random sound field environment that do not fit in a standard model
 - e.g. for DFAT, one could establish the sound field with a rigid model, characterize it with FSP or GSP and apply it to the test article.
- GSP enables **continuous FE–SEA workflows from the same dataset**
- **Engineering implication:** Using TBL in separated flow regions can lead to non-conservative load predictions
- For complex geometries with flow separation, CFD-based methods (FSP/GSP) provide improved correlation
- FSP is still needed:
 - Where phase matter
 - To account for local environments



Thank you

